

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	K.Dhilip
Register Number	192365013

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

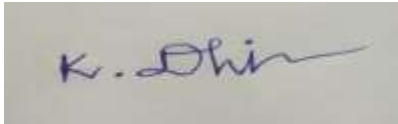
Duration : 60 Minutes

Code of Conduct

I **K.Dhilip/192365013** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

AIM: To Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

step1: Go to zoho.com.

step 2: Log into the zoho.com.

step 3: Select one application step.

step4: Enter application name as SIMATS Library Management System.

step 5: Created new application Library Management System.

step 6: Select one form

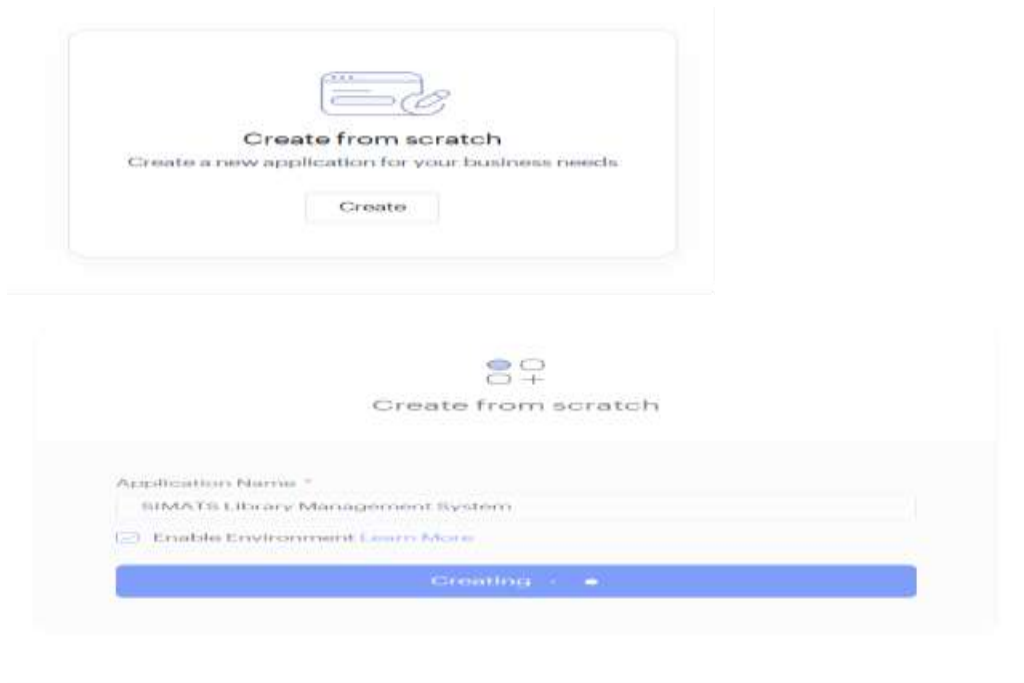
step 7: The software has been created

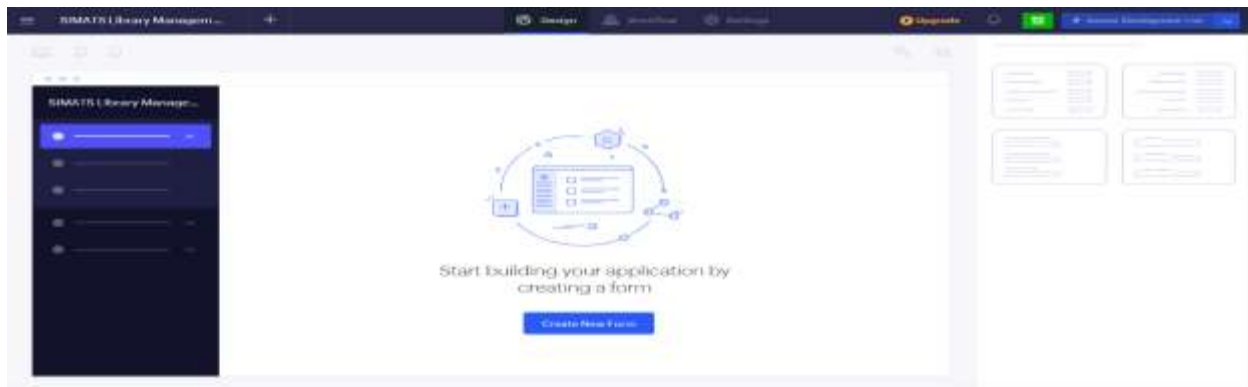
Implementation

Step 1:



Step 2:





Step 3:

Book Details

Title of the Book:

Author:

Publication:

Year:

Edition:

No of Copies:

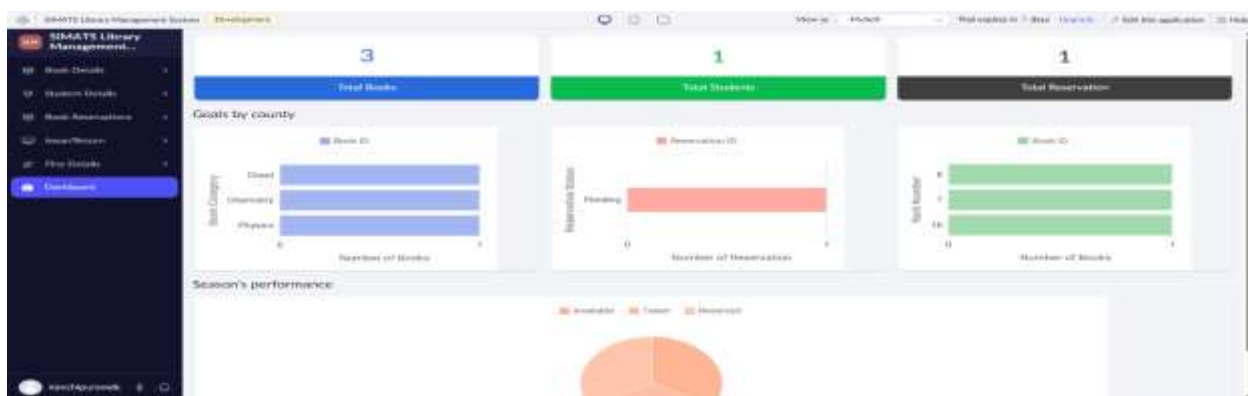
Rack No:

Category:

Status:

Book Details Report

Book ID	Title of the Book	Author	Publication	Year	Edition	No of Copies
2	Physics	Dr. Khosro	Yes	1955	5	10
1	Cloud	Bob	Yes	1987	5	3



Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

AIM: To Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

step1: Go to zoho.com.

step 2: Log into the zoho.com.

step 3: Select one application step.

step4: Enter application name as Payroll Management System.

step 5: Created new application Payroll Management System.

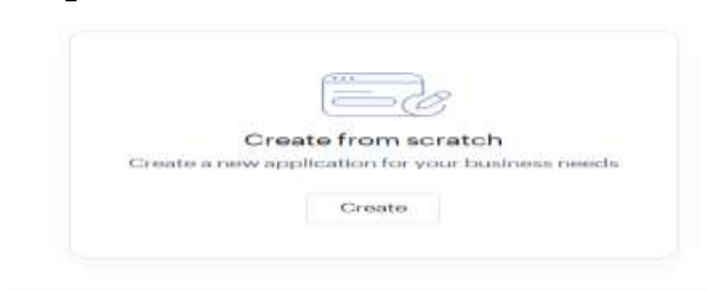
step 6: Select one form

step 7: The software has been created

Step 1:



Step 2:



Step 3:



Step4:

Create from Scratch

Name form

Create Form

Create from Scratch

Name form

Create Form

Create from Scratch

Name form

Create Form

Step 7:

Run workflow on any event in the form

Choose form

Payroll Processin



Run when a record is
(Record Event)



Created



Edited



Created or Edited



Deleted

When to trigger workflow
(Form Event)

User Input of a field



Choose Field

Basic Pay



Name the workflow

calc basic pay

Creating •

Step 8:

Create from Scratch

Name form

Payslip / Payment Details

Create Form

Step 9:

Smart Payroll Management...

Employee Details

Employee Details

All Employees Details

Salary Details

Payroll Process

Payable / Payment Details

Dashboard

Employee Details

Employee Name

Chirp K

Department

IT

Designation

Head of all

Date of Joining

17-Jan-2023

Experience in Present Organization

4

Overall Experience

10

Phone Number

+91 1234567891

Email

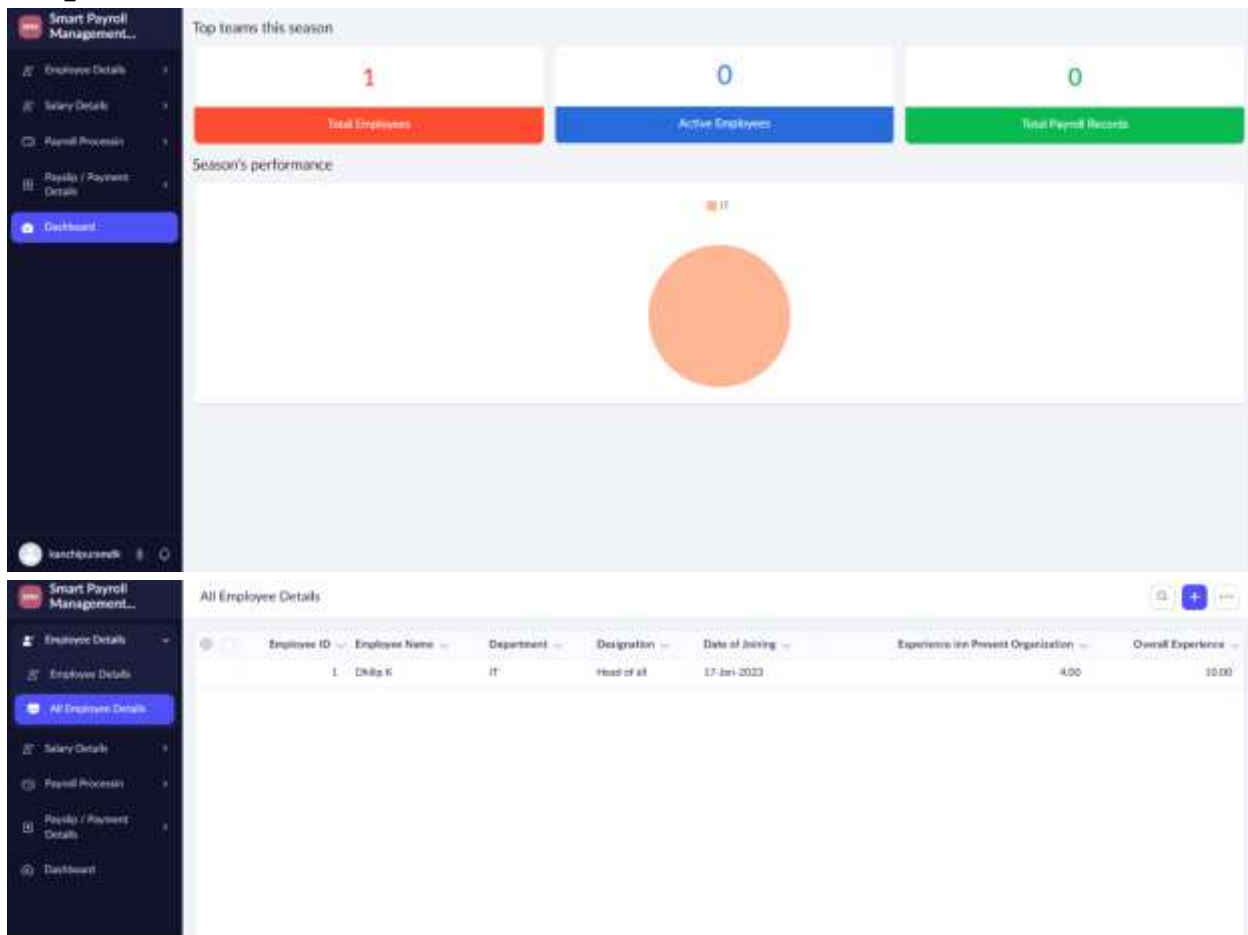
ch@gmail.com

Employment Status

Active

Submit

Step 10:



Smart Payroll Management...

Employee Details

Salary Details

Payroll Process

Payable / Payment Details

Dashboard

All Employee Details

Employee ID	Employee Name	Department	Designation	Date of Joining	Experience in Present Organization	Overall Experience
1	Chirp K	IT	Head of all	17-Jan-2023	4.00	10.00

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

AIM:

To demonstrate virtualization by installing type-2 hypervisor in your device, create and configure VM image with a host operating system (either windows/linux).

PROCEDURE:

STEP 1:Download VMware workstation and installed as type 2hypervisor.

STEP2:Download ubuntu or tiny OS as iso image file.

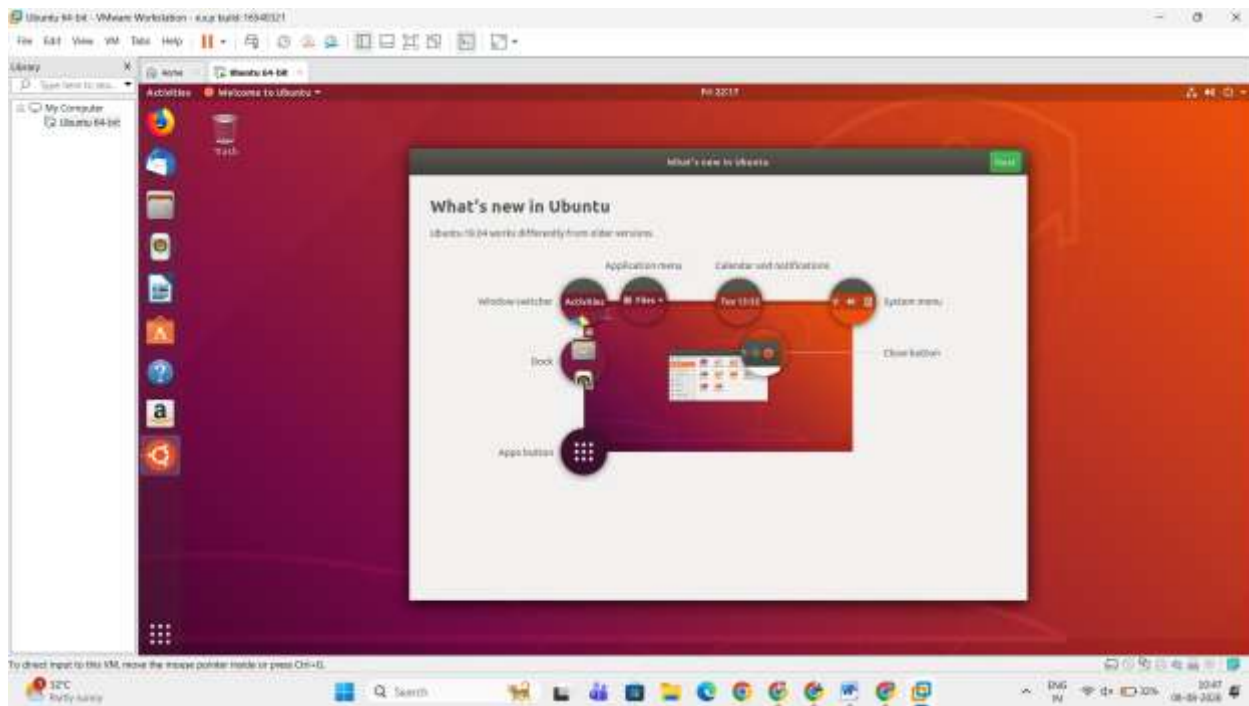
STEP 3: In VMware workstation->create new VM.

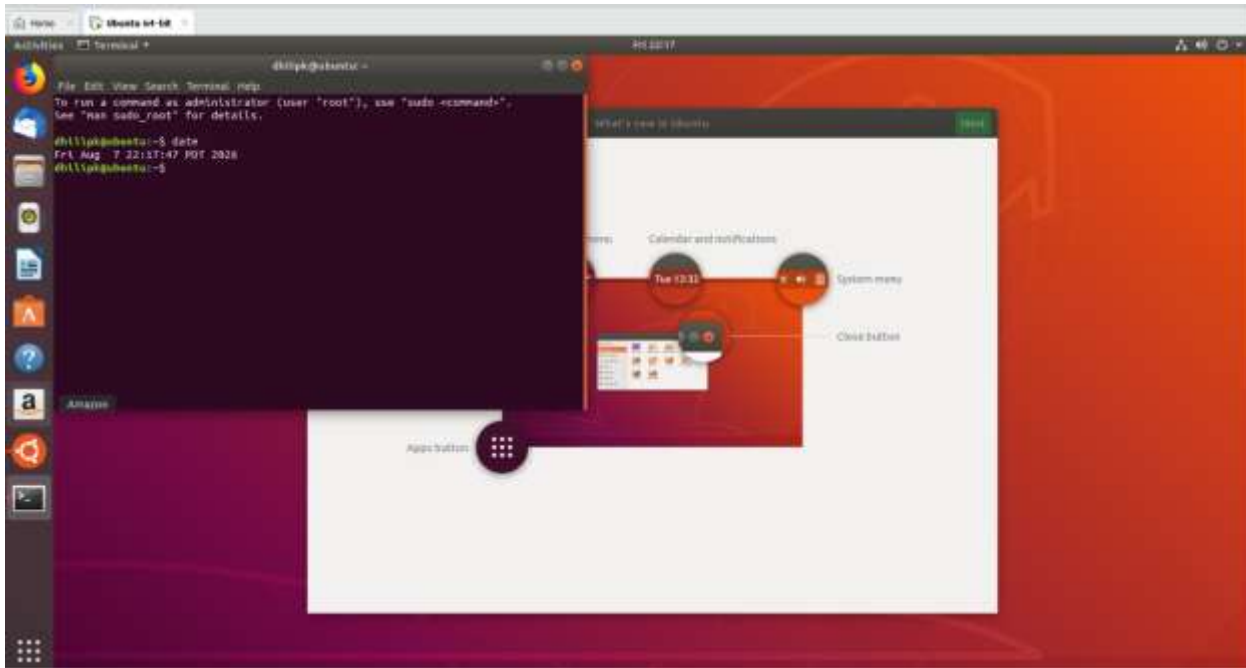
STEP 4: Do the basic configuration settings.

STEP 5: Created tiny OS virtual machine.

STEP 6: Launch the VM.

Implementation:





Result:

The Ubuntu virtual machine was successfully created using VMware Workstation.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

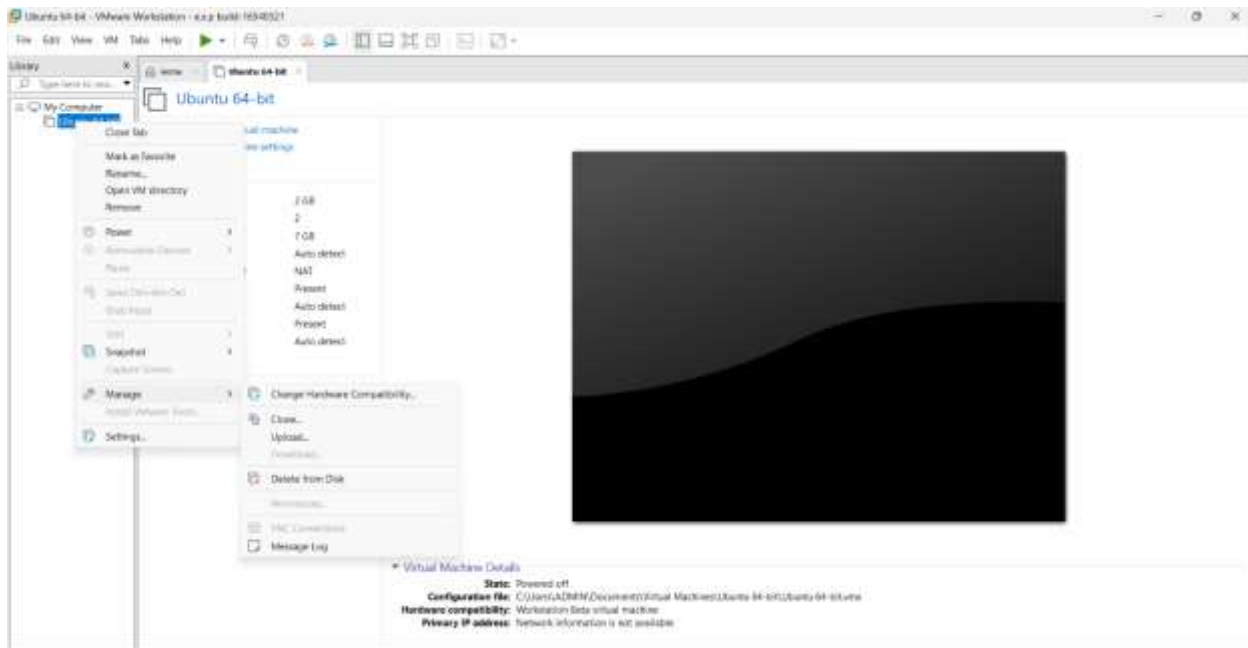
Aim

To demonstrate virtualization using **VMware Workstation (Type-2 Hypervisor)** by creating a virtual machine, cloning it, and testing the cloned VM.

Procedure

1. Install and open **VMware Workstation** on the Windows host system.
2. Create a new virtual machine and install **Ubuntu Linux** as the guest OS.
3. Configure the VM with suitable RAM, processor, storage, and network settings.
4. Shut down the original VM and create a copy of its VM folder.
5. Add the copied VM to VMware Workstation and name it **Ubuntu Clone**.
6. Select **"I Copied It"** when VMware asks about the VM identity.
7. Start the cloned VM and verify that Ubuntu and the previous files/configuration are loaded successfully.

Implementation:



Clone Virtual Machine Wizard

Clone Source

Which state do you want to create a clone from?

Clone from

☒ The current state in the virtual machine

Creating a linked clone from the current state will create a new snapshot.

☐ An existing snapshot (powered off only):

This virtual machine has no existing cloneable snapshots.

< Back

Next >

Cancel

Clone Virtual Machine Wizard

Clone Type

How do you want to clone this virtual machine?

Clone method

☒ Create a linked clone

A linked clone is a reference to the original virtual machine and requires less disk space to store. However, it cannot run without access to the original virtual machine.

☐ Create a full clone

A full clone is a complete copy of the original virtual machine at its current state. This virtual machine is fully independent, but requires more disk space to store.

< Back

Next >

Cancel

Clone Virtual Machine Wizard

Name of the New Virtual Machine

What name would you like to use for this virtual machine?

Virtual machine name

Clone of Ubuntu 64-bit

Location

C:\Users\ADMIN\Documents\Virtual Machines\Clone of Ubuntu

Browse...

< Back

Finish

Cancel

